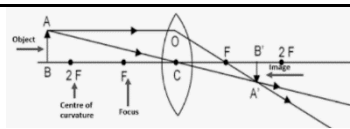
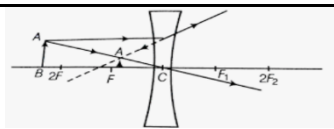
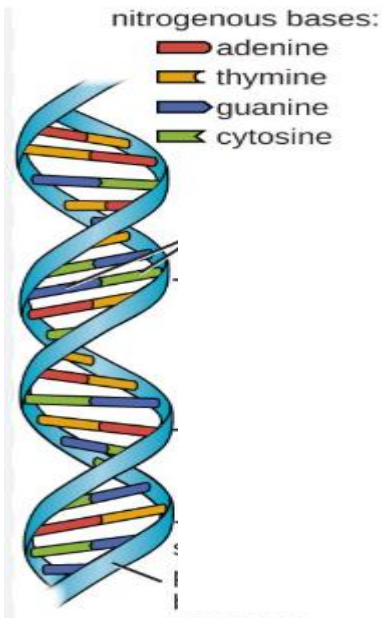
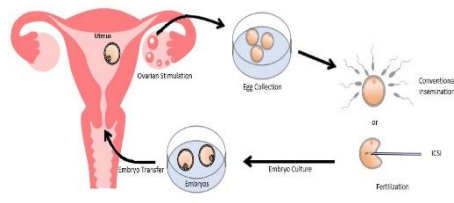
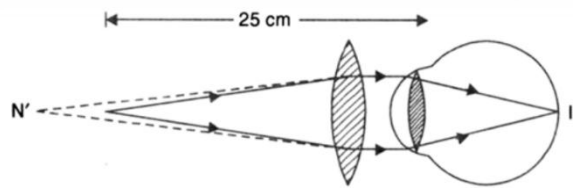


SEE 2082 (2026)
अनिवार्य विज्ञान तथा प्रविधि
उत्तरकुञ्जिका

यस कुञ्जिकामा भएको बाहेक अन्य सही वैकल्पिक तरिकाबाट समस्या समाधान गरेमा पनि अंक प्रदान गर्नुपर्ने छ ।
 उत्तरकुञ्जिकामा प्रत्येक चरणको प्राप्ताङ्क १ प्रदान गर्नुपर्नेछ तथा एकाइ छुट भएमा ०.५ अङ्क मात्र घटाउनुपर्ने छ ।

प्र. नं.	उत्तर	अंक
खण्ड 'क' (Section 'A')		
1 a.	(ii) Chatting on the internet	1
b	(iii) Algae	1
c	(ii) Pulmonary artery	1
d	(iv) Turtle	1
e	(iii) 175 N	1
f	(ii) Due to up thrust exerted by water on plastic bottle	1
g	(iv) x- red, y- violet	1
h	(iii) Big bang theory	1
i	(iii) Pentyne	1
j	(iv) Bromadiolan, cumatetralyl, zinc phosphate	1
खण्ड 'ख' (Section 'B')		
2.a	-Solubility of sugar	1
b	- Earthworm → Cockroach → Fish → Frog	1
c	Pressure in arteries during heart contraction	1
d	- Pascal's law- Pressure applied to a confined fluid spreads equally in all directions.	1
e	- $p = 1/f$ $f = 1/p = 1/4 = 25 \text{ cm or } 0.25 \text{ m}$	1
f	 - or 	1
g	- 50 Hz AC means the current repeats its pattern 50 times every second.	1
h	- Flat Universe	1
i	- $\text{Mg} + 2\text{H}_2\text{O} \rightarrow \text{Mg}(\text{OH})_2 + \text{H}_2 \uparrow$	1
खण्ड 'ग' (Section 'C')		
3	- $S=ut+1/2a^2t$ - Unit of s is m - Unit of ut is $\text{m/s} \times \text{s} = \text{m}$ - Unit of a^2t is $(\text{m/s}^2)^2 \times \text{s} = \text{m}^2/\text{s}^3$ so, the relation is incorrect.	1
4	- Thallus body structure - asexual reproduction They are amphibious plants or any others related	1

5	- Even though males have XY chromosomes , whether children are sons or daughters depends on which sperm (X or Y) fertilizes the egg , so some may have only sons or only daughters by chance.	1 for each									
6	- (a) Meiosis - (b) Mitosis	1 mark for each									
7	- (1-3 days)- Cleans the cells of the hive or Feeds larvae and queen . - (18-20days)- Collects nectar, pollen, water, and propolis .	1 1									
8	- Yarsagumba - Protect the high-altitude alpine meadows where Yarsagumba grows. or others	1 1									
9	- An object's weight is slightly more at the base of a mountain than at its top. - Reason- The base is closer to the center → gravity is stronger → weight is slightly more	1									
10	- $F = G \frac{m_1 \times m_2}{d^2}$ - $m_1 = 1\text{kg}$, $m_2 = 1\text{ kg}$ and $d = 1\text{m}$	1 1									
11	- use the $\frac{V_p}{V_s} = \frac{N_p}{N_s}$ relation and calculate $\frac{N_p}{N_s}$ - - - Ans 44:1	1 1									
12	- Increasing the number of turns on coil - Increasing the strength of magnet or any other related	1 1									
13	$2H_2O_2 \xrightarrow{MnO_2} 2H_2O + O_2 \uparrow$ Catalyst: Manganese dioxide (MnO_2) or any other correct answer	1 1									
14	- Anode- pure metal but correct is impure metal and Cathode- impure metal but correct is pure metal - Electrolyte – $CuSO_4$ incorrect but correct is $AgNO_3$	1 1									
15	<table border="1"> <thead> <tr> <th>Feature</th><th>Alkane</th><th>Alkyne</th></tr> </thead> <tbody> <tr> <td>Number of bonds between carbon atoms</td><td>Only single bonds (C–C)</td><td>At least one triple bond (C≡C)</td></tr> <tr> <td>General molecular formula</td><td>C_nH_{2n+2}</td><td>C_nH_{2n-2}</td></tr> </tbody> </table>	Feature	Alkane	Alkyne	Number of bonds between carbon atoms	Only single bonds (C–C)	At least one triple bond (C≡C)	General molecular formula	C_nH_{2n+2}	C_nH_{2n-2}	1 1
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16	- Safer for health - preserves natural taste and nutrients. Or any other relate	1 1									
खण्ड “घ” Section “D”											
17	(a) Instagram and what's app and any others related (b)- Create study groups to discuss topics. -Share notes, study materials, and useful links .	1+1 =2 1+1 = 2									

18	(a) X (b) Y (c) Z (d) Y	1 for each 1 1 1
19	(a) A because of high specific heat capacity, $Q \propto S$ for constant mass 2000g and constant temperature difference 20°C . (b) Use $Q = ms\Delta t$ and find s for c then again use $Q = ms\Delta t$ find m using $s = 385\text{J/kg}^{\circ}\text{C}$ and obtain from above, $m = 2.38\text{ kg}$	$1 + 1 = 2$ $1 + 1 = 2$
20	(a)  nitrogenous bases: █ adenine █ thymine █ guanine █ cytosine or other correct diagram (b) IVF is a technique in which an egg and sperm are combined outside the body (in a lab) to form a zygote, which is then implanted into the uterus to achieve pregnancy. 	4
21	(a) Hypermetropia or far (long) sightedness (b) Correct fig  (c) Concave lens	1 2 1
22	(a) B -metal and X - Nonmetal (b) Zero (c) MgCl_2 (Magnesium chloride)	$1 + 1 = 2$ 1 1
23	(a) Carbon dioxide (CO_2) (b) Calcium carbonate (CaCO_3) and Dilute hydrochloric acid (HCl) (c) reaction $6\text{CO}_2 + 6\text{H}_2\text{O} \xrightarrow{\text{light, chlorophyll}} \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ (d) Blue to red	1 1 1 1